Carbon Blade: Carbon Capture Device

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Project Overview

- Increasing atmospheric carbon dioxide is a key driver of global climate change. Reduction in CO_2 is necessary to minimize climate impacts.
- This project sought to design a **Passive Direct Air Contactor** (DAC) with low energy demands.
- The DAC's blades are coated with a **sodium hydroxide solvent** which will react with CO_2 in the air, pulling it out of the atmosphere. (Fig. 1).

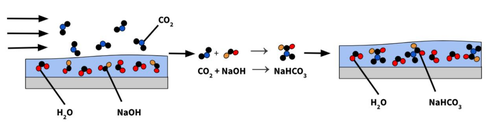


Figure 1: Chemical Reaction

- The sodium bicarbonate product is then fed into a secondary process to recover purified CO_2 for other applications.

Our Design Solution

- **Functional Requirements:**
  - **0.1 metric tons/day** of CO_2 capture.
  - **100 mL/min** of solvent flow rate over the blades.
  - **Capture efficiency of 50%** of the CO_2 passing through the DAC.
  - Blade stack **RPM of 1**.
  - Minimized power consumption.

Required Surface Area Analysis: Ensures adequate amount of surface area for diffusion of CO_2 into NaOH.

$$A_{Req} = \delta \frac{\dot{m}_{co_2}}{D_{co_2}} = 0.3 \frac{1.157\ g/s}{1.8 mg/s/m^2 (10^{-3})} = 192.83\ m^2$$

Helix Blade Length Analysis: Ensures helix with a certain radius will have the required surface area:

$$SA_{helicoid} = \pi N\{(r_h)\sqrt{(\frac{p}{2\pi})^2 + (r_h)^2} + (\frac{p}{2\pi})^2 ln(r_h) + \sqrt{(\frac{p}{2\pi})^2 + (r_h)^2})\} = 205.46\ m^2$$

$$r_h = 0.76\ m,\ p = .0762\ m,\ N = 100\ turns$$

Our Design Solution

- A **Cascading Helix DAC** was designed to fit the functional requirements and maximize flow behaviors that encourage CO_2 capture.

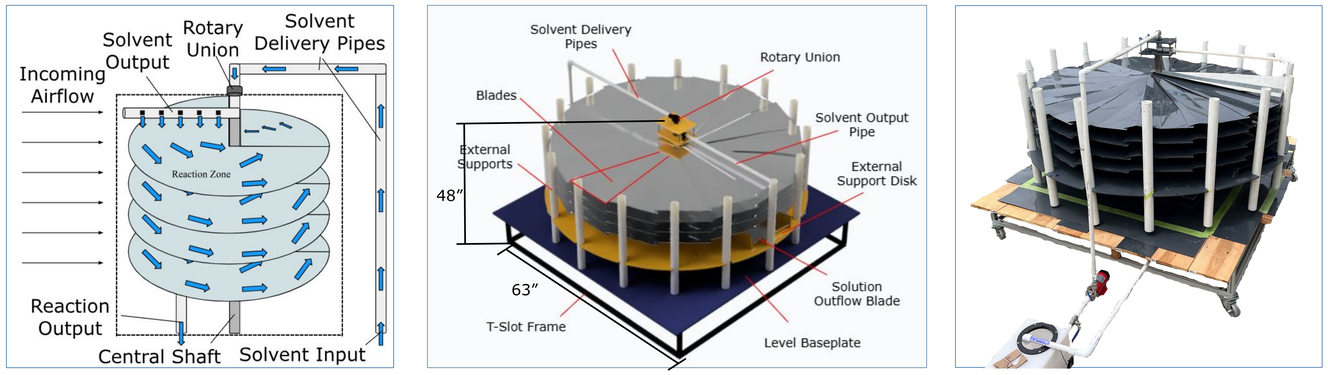
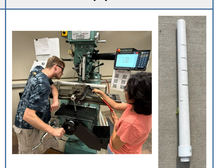


Figure 2: Cascading Helix DAC flow diagram, CAD, and final prototypes with labels

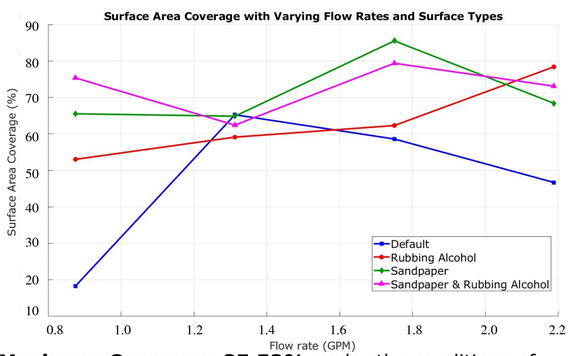
Main Components		
Blades: • Serve as surface for CO_2 diffusion with solvent. • Angled to promote fluid spreading. • Square shaft connections with four configurations.	**Solvent Delivery:** • Pumped up to the side of assembly (static). • Rotary union serves as static to rotating transition. • Irrigation-style holes to promote uniform solvent delivery.	**External Supports:** • Ensure blades remain level. • Multiple slots, each support holds four blades. • Attached to central shaft via support plate of ¼" PVC and steel struts.

Prototype Manufacturing

CNC Router: PVC Components	Mill & Lathe: Metal Adapters	Heat Gun: Blade Bending
Waterjet: External Support Disk	**Mill & Welding: Motor Box**	**Mill: External Supports**
		

Prototype Testing

Testing goal: Determine ideal flow rate and surface conditions for maximum solvent coverage surface area.



Maximum Coverage: 85.52% under the conditions of sandpaper surface treatment and 1.75 GPM flow rate.

Conclusions

- **Surface treatments** significantly improve solvent coverage.
- Excessive flow rates limit spreading and increase spillage.
- PVC blades deflect under their own weight leading to **backflow**.
- Nozzle orifice design lead to uneven solvent distribution, compounding to flow patterns.

Future Work

- Creating a **full size prototype** with **stainless steel blades** with **surface features** to improve performance and reach desired capture rate.
- Confirmation of **true capture rate** of the DAC.
- **Assembly improvements** & simplifications for full scale production.

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